

Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
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Roll Number	2520030138
Branch	CSE
Course Outcome	6

Case Study Topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform

Work Done Summary:

Developed an automated data-cleaning module for smart city networks using the Levenshtein Edit Distance algorithm. The platform intercepts raw, corrupted operational commands sent over noisy wireless channels by IoT sensors, comparing them against verified database command schemas to determine the minimum number of character-level modifications (insertions, deletions, and substitutions) needed for rectification. Dynamic Programming was employed to reliably compute the cost matrix without redundant structural analysis. The solution serves as an inline error-filtering layer that prevents system faults across central dashboard consoles, traffic control units, and automated smart-grid switches.

Implementation Details:

- Defined corrupt inputs and matching command string candidates within the processing engine.
- Constructed a two-dimensional Dynamic Programming table for step-by-step cost computation.
- Defined standard localized penalties for character insertions, deletions, and substitutions.
- Retained solved sub-problem states using a bottom-up tabulation strategy.
- Extracted the exact minimum edit distance value from the final cell of the matrix.
- Evaluated string structures sequentially character by character.
- Displayed full diagnostics showing corrupt strings alongside recommended structural corrections.
- Validated pipeline scaling characteristics against standard operational big-O boundaries.

Result: Screenshots / Output

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=====
      URBANAI - IoT SENSOR COMMAND CORRECTION SYSTEM
=====

Received Sensor Command : status_chck
Valid Database Schema   : status_check

Minimum Edit Distance = 1

Suggested System Correction:
status_check

Complexity Analysis:
Time Complexity   : O(m * n)
Space Complexity  : O(m * n)

Command Integrity Validation Completed Successfully.
```

Time Complexity:

- Time Complexity : $O(m \times n)$
- Space Complexity : $O(m \times n)$
-

Conclusion:

The UrbanAI IoT Sensor Command Error Correction System utilizing Edit Distance successfully identified transmission anomalies and restored corrupt inputs to standard structural formats. Dynamic Programming minimized computation overhead, enabling fast inline string sanitization. This work demonstrates the crucial value of DP-based string alignment algorithms in securing data integrity across messy, distributed IoT smart city networks.

GitHub Link:

<https://github.com/marahrudaydeepak-ctrl/DSA-2>

Student Signature	Faculty Signature